Lecture 3. Cholesky Decomposition

Existence and uniqueness, algorithm and cost, pivoting, applications

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Mantra 2

The structure should be exploited whenever possible in solving a problem.

Common structures are symmetry, positive (semi)definiteness, sparsity, Kronecker products and low rank. Statisticians mostly meet positive (semi)definite matrices: covariance matrices and Gram matrices $X^\top X$. A matrix M is positive definite (semidefinite) if $x^\top Mx > 0$ (≥ 0) for all $x \neq 0$; since $x^\top Mx = x^\top \frac{M+M^\top}{2} x$, we may assume M symmetric.

3.1 The Cholesky decomposition

Theorem 3.1 (Cholesky)

If $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite, then $A = LL^\top$ for a unique lower triangular L with positive diagonal entries.

Proof. Induction on n. For $n = 1$, $a = \sqrt{a}\sqrt{a}$ with $a > 0$. For $n > 1$, partition and match blocks:

$$\begin{pmatrix} a_{11} & \mathbf{a}^\top \\ \mathbf{a} & A_{22} \end{pmatrix} = \begin{pmatrix} \ell_{11} & \mathbf{0}^\top \\ \mathbf{b} & L_{22} \end{pmatrix} \begin{pmatrix} \ell_{11} & \mathbf{b}^\top \\ \mathbf{0} & L_{22}^\top \end{pmatrix} \iff a_{11} = \ell_{11}^2, \quad \mathbf{a} = \ell_{11}\mathbf{b}, \quad A_{22} = \mathbf{b}\mathbf{b}^\top + L_{22}L_{22}^\top.$$

Since $a_{11} = e_1^\top A e_1 > 0$, the conditions force $\ell_{11} = \sqrt{a_{11}}$ and $\mathbf{b} = \mathbf{a}/\sqrt{a_{11}}$. The **Schur complement** $S = A_{22} - \mathbf{b}\mathbf{b}^\top = A_{22} - \mathbf{a}\mathbf{a}^\top/a_{11}$ is positive definite: for $\mathbf{y} \neq \mathbf{0}$, put $x = (-\mathbf{a}^\top\mathbf{y}/a_{11}, \mathbf{y}^\top)^\top \neq 0$; then

$$0 < x^\top Ax = \frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} - 2\frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} + \mathbf{y}^\top A_{22}\mathbf{y} = \mathbf{y}^\top S\mathbf{y}.$$

By the induction hypothesis $S = L_{22}L_{22}^\top$ for a unique such L_{22}, which completes existence and uniqueness. ■

The proof is the algorithm: compute the first column of L, update the Schur complement, and recurse; this can be done in place in the lower triangle of A.

Cost. The Schur complement updates are symmetric, so only half the entries are computed:

$$\tfrac{1}{2}\{2(n-1)^2 + 2(n-2)^2 + \cdots + 2\} \approx \tfrac{1}{3}n^3 \text{ flops}$$

plus n square roots, *half* the cost of LU. The algorithm is very stable without pivoting, and it fails (a nonpositive number under a square root) exactly when A is not positive definite. So Cholesky is the cheapest way to *test* positive definiteness.

Pivoting for semidefinite matrices. If A is only positive semidefinite (e.g. $X^\top X$ with X rank deficient), a zero pivot can occur. With **symmetric pivoting**, at stage k the largest remaining diagonal entry is moved (by a symmetric row and column swap) to the pivot position. If the largest remaining diagonal entry is 0, the whole remaining block is 0: a positive semidefinite matrix with zero diagonal entry a_{ii} has zero row i, since $|a_{ij}| \leq \sqrt{a_{ii}a_{jj}}$. The algorithm then stops, giving

$$PAP^\top = LL^\top, \qquad L \in \mathbb{R}^{n\times r}, \quad r = \operatorname{rank} A.$$

LAPACK provides `potrf` (no pivoting) and `pstrf` (pivoting); R's `chol` returns the upper factor $R = L^\top$, $A = R^\top R$.

3.2 Applications

Multivariate normal log-likelihood. For $y \sim N_n(0, \Sigma)$,

$$\log f(y) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\log\det\Sigma - \frac{1}{2}y^\top \Sigma^{-1} y.$$

- Method 1: invert Σ ($2n^3$), form the quadratic form ($2n^2$), compute the determinant by LU ($\frac{2}{3}n^3$).
- Method 2: factor $\Sigma = LL^\top$ ($\frac{1}{3}n^3$), solve $Lx = y$ by forward substitution (n^2), then $y^\top\Sigma^{-1}y = x^\top x$ ($2n$) and $\log\det\Sigma = 2\sum_i \log \ell_{ii}$ (n).

Method 2 is about 8 times cheaper and more accurate, and the log-determinant computed this way cannot overflow. To evaluate the density at many points, factor once and pay n^2 per point.

Linear regression. For $X \in \mathbb{R}^{n\times p}$, solve the normal equations $X^\top X\beta = X^\top y$ by Cholesky:

form $X^\top X$	np^2 (symmetric)
form $X^\top y$	$2np$
Cholesky of $X^\top X$	$\frac{1}{3}p^3$
two triangular solves	$2p^2$
$(X^\top X)^{-1}$ for standard errors	$\frac{4}{3}p^3$

In total about $np^2 + \frac{1}{3}p^3$ flops, or $np^2 + \frac{5}{3}p^3$ with standard errors. This is the fastest method when $n \gg p$, but forming $X^\top X$ squares the condition number (Lecture 4).